

Exploring Transaction Flow in Hyperledger Fabric

This is a continuation of the article that introduced Hyperledger Fabric, published in the previous issue of *OSFY*. It explores the internal workings of Hyperledger Fabric and takes a look at transaction flow.



In a distributed ledger system, consensus is the process of reaching an agreement on the next set of transactions to be added to the ledger. In Hyperledger Fabric, consensus involves three distinct steps:

- Transaction endorsement
- Ordering
- Validation and commitment

These three steps ensure the policies of a network are upheld. We will look at how these steps are implemented by exploring the transaction flow.

Transaction flow (Step 1)

Within a Hyperledger Fabric network, transactions start out with client applications sending transaction proposals or, in other words, proposing a transaction to endorsing peers.

Client applications are commonly referred to as applications or clients, and allow people to communicate with the blockchain network. Application developers can leverage

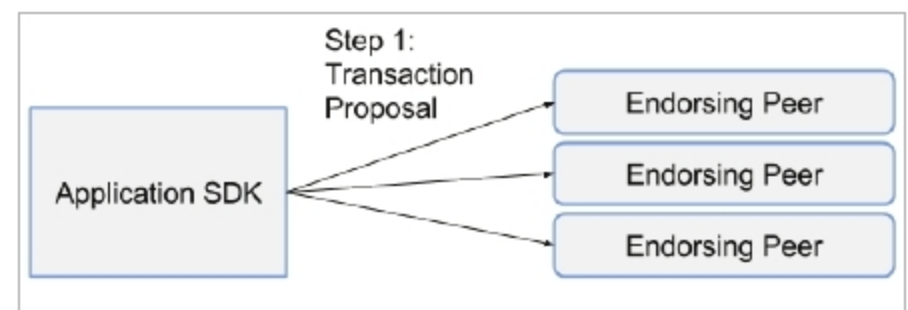


Figure 1: Step 1 of the transaction flow

the Hyperledger Fabric network through the application's SDK.

Transaction flow (Step 2)

Each endorsing peer simulates the proposed transaction, without updating the ledger. The endorsing peers will capture the sets of read and written data, called RW sets. These RW sets capture what was read from the current world state while simulating the transaction, as well as what would have been written to the world state, had the transaction been executed.